

Yixiong Sun

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PROFESSIONAL SUMMARY

Systems neuroscience PhD candidate and software engineer combining *in vivo* electrophysiology, machine learning, signal processing, and computational modeling to analyze noisy, high-dimensional neural data. Experienced leading end-to-end research from experimental design and data acquisition through modeling, validation, and scientific communication, with prior production software experience spanning full-stack, mobile, cloud and DevOps systems.

EDUCATION

University of Toronto Toronto, ON
PhD, Cell and Systems Biology (Systems Neuroscience) Sep 2021 - Present
Thesis: Linking abnormal hippocampal neural activity and memory deficits to A β accumulation in Alzheimer's disease
GPA: 4.0/4.0

McGill University Montreal, QC
Bachelor of Science, Joint Honours Computer Science and Biology Sep 2017 - Apr 2021
GPA: 3.96/4.0

TECHNICAL SKILLS

Programming: Python, C/C++, Objective-C, Swift; Java, JavaScript, Git, Linux, Bash
Machine Learning and AI: PyTorch, scikit-learn, NumPy, pandas, SciPy, supervised and representation learning, CNNs, model evaluation, uncertainty quantification, LLM-assisted extraction and retrieval
Signal and Scientific Computing: time-series analysis, digital signal processing, spectral analysis, event detection, graph analysis, statistics, computational modeling, Brian2
Software and Cloud: Docker, CI/CD, AWS, Node.js, Django, PostgreSQL, MongoDB, Redis, APIs, testing, production/staging infrastructure
Neuroscience: *in vivo* electrophysiology, spike/LFP analysis, closed-loop stimulation, behavioral experiments, neural data quality control

EXPERIENCE

Takehara Lab, University of Toronto Toronto, ON
PhD Candidate, Systems Neuroscience Sep 2021 – Present

- Led a multi-year research program on hippocampal network dynamics and memory in Alzheimer's disease, integrating *in vivo* electrophysiology, behavior, signal processing, computational modeling, and statistics.
- Built reproducible Python and MATLAB pipelines to process and analyze approximately 4 TB of multichannel electrophysiology, behavioral, and imaging data across over 30 experimental subjects, automating event detection, population analyses, statistical testing, and visualization.
- Developed deep-learning and signal-processing workflows to classify sharp-wave ripples, interictal epileptiform events, and noise; evaluated cross-subject generalization through subject-held-out validation and architecture and loss-function ablations
- Built and validated mechanistic neural-network simulations against experimental observations, translating biological hypotheses into quantitative model comparisons and falsifiable predictions.
- Designed and executed *in vivo* electrophysiology, behavioral, and closed-loop stimulation experiments; established quality-control procedures and coordinated workflows from data acquisition through analysis.
- Authored two manuscripts and presented seven posters and three talks at neuroscience and data-science venues, including the Society for Neuroscience and Canadian Association for Neuroscience.

Waldispühl Lab, McGill University Montreal, QC
Undergraduate Researcher Sep 2020 – Apr 2021

- Applied and extended the BayesPairing2 RNA secondary-structure and motif discovery pipeline; contributed to validation and optimized Bayesian-network construction and RNA model loading.

Beagle Inc Kitchener, ON
Software Developer May 2021 – Aug 2021

- Improved PDF/Word document parsing and NLP pipelines for legal phrase recognition using tokenization and classifiers including SVMs, random forests, and multilayer perceptrons; built a lightweight continuous-learning labeling workflow.

- Refactored and containerized a legacy Django platform and built CI/CD workflows for deployment to load-balanced AWS EC2 instances with S3 storage and PostgreSQL on Amazon RDS.
- Improved testing and documentation and demonstrated the relaunched platform to investors.

iVenuto.com Corporation

Software Developer

Kitchener, ON

May 2018 – Aug 2021

- Built and deployed full-stack features for a visitor-management platform, Zap In, including WebRTC video calls, Google Dialogflow speech input, emergency workflows, and offline records.
- Established testing and CI/CD workflows; maintained staging and production infrastructure using AWS EC2/S3, Redis, and MongoDB; supported iOS testing and released over 10 versioned updates.
- Led interns by designing features and supervising implementation; resolved escalated technical issues for approximately 50 customers over 4 years.

GHP Group Inc.

Mobile Software Developer

Remote

Jan 2016 – Dec 2016

- Built and released iOS and Android applications for Bluetooth Low Energy devices, implementing BLE pairing plus bidirectional command and response handling.

SELECTED TECHNICAL PROJECTS

EvidNeuro – Evidence-Grounded Neuroscience Retrieval Engine

Aug 2026 – Present

TypeScript, LLM-assisted extraction, MinerU, BM25, vector search, pgvector, PostgreSQL, knowledge graphs, Allen Brain Map, UBERON, Cell Ontology

- Built a scientific retrieval engine that converts research PDFs into layout-aware evidence objects, typed claims, measurements, normalized entities, and directed knowledge-graph relations, with provenance linking every result to its source.
- Implemented hybrid BM25/vector retrieval with reciprocal-rank fusion and optional LLM reranking, plus species-aware ontology normalization, evidence-strength grading, cross-paper contradiction detection, and cross-scale graph queries.
- Validated the system with 1,017 passing tests across 92 test files and ingested a 60-paper neuroscience corpus containing 15,170 evidence objects, 9,047 claims, 598 measurements, and 2,990 directed relations.

Multimodal Neural Event Classification

Jan 2026 - Present

Python, PyTorch, signal processing, CNNs, subject-aware evaluation

- Built an end-to-end pipeline to detect, curate, and classify neural events as sharp-wave ripples, interictal epileptiform discharges (IEDs), or noise across 7,701 labeled events from 30 subjects.
- Developed a multimodal CNN combining LFP waveforms, time-frequency spectrograms, cross-channel attention, and coherence and entropy-based features.
- Conducted leakage-free, subject-held-out cross-validation and paired architecture/loss ablations, achieving 86.3% accuracy and 0.859 macro F1 across three folds, substantially outperforming a naïve thresholding baseline of 60.8% accuracy and 0.313 macro F1.

SELECTED PUBLICATIONS AND PRESENTATIONS

Sun Y, Chekhov S, Zhang K, Margarian S, Bogle P, Han CJ, Ando K, Suzuki T, Fraser PE, Taxidis J, Takehara-Nishiuchi K. Sharp wave-ripple disruption limits experience-dependent neuronal ensemble reorganization in amyloid-associated memory impairment. *Current Biology*, 2026. *In review*.

Sun Y, Takehara-Nishiuchi K. The medial prefrontal cortex leaves the hippocampus when it prepares for the future. *Science Progress*, 2024.

Sun Y, Taxidis J, Takehara-Nishiuchi K. In silico modeling of amyloidosis-induced CA1 spike abnormalities: Unraveling the role of excess synaptic glutamate. *Society for Neuroscience*, 2025.

AWARDS

Data Sciences Institute Doctoral Fellowship - \$25 000/year

Sep 2024 – Present

University of Toronto Fellowship

Sep 2021 – Aug 2026

Dr. Feng Qian Scholarship - \$3000

Sep 2019

JW McConnell Scholarship - \$1500

Sep 2017